

L-Università ta' Malta

# Machine Learning Approaches to the Blockchain

*some hyped-up tagline*


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Co-supervised by Dr Who

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Department of Alchemy
University of Malta


**October 2026**

*A dissertation submitted in partial fulfilment of the requirements for the degree of M.Sc. in Your Degree.*




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WWW.UM.EDU.MT

*First edition, 26 October 2026*

*To The Avengers*

*You know, for saving the world.*

# Acknowledgements

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# Abstract

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# List of Abbreviations

**CDMA** Code Division Multiple Access . . . . . 10
**GSM** Global System for Mobile communication . . . . . 10
**TDMA** Time Division Multiple Access . . . . . 10
**UA** Used Acronym . . . . . 10

# 1

# Introduction

Note that you may have multiple `\include` statements here, e.g. one for each subsection.

General structure of this chapter should read as follows. This chapter should be used to motivate your study and answer the question "Why is this important?" Also, it should define what you set out to achieve (these will be revisited in the conclusions chapter). You should describe your approach to the Aims and Objectives, including an evaluation part.

## 1.1 | Motivation

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## 1.2 | Aims and Objectives

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## 1.3 | Our Approach

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## 1.4 | Document Structure

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# 2

# Background & Literature Review

In this section you need to explain all the theory required to understand your dissertation (i.e. the following chapters). But really in this chapter I am going to show you some examples.

## 2.1 | Some Technique One

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### 2.1.1 | Some Sub-technique One

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#### 2.1.1.1 | Some Sub-sub-technique One

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## 2.2 | Some Technique Two with Super Long Title Which Will Overrun In Header

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Imagine some colourful description on Some Technique Three.

## 2.3 | Evaluation Criteria

This section should contain information on the metrics and background used to evaluate your work.

## 2.4 | Related Work

**In this section you need to explain (and reference) similar work in literature**. Make sure to:

- Give a systematic overview of papers with related/similar work
- Highlight similarities/differences to your work (perhaps in the form of a table)

Note that this section may be sectioned based on the different aspects of your dissertation.

## 2.5 | Summary

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# 3

# Materials & Methods

This section should include a recipe of what you did (explain what you have done so if someone wants to reproduce the experiment, they can). A flow chart is typically helpful. Also, make sure to define all software that you used including version numbers and OS. Should also include a description of statistical methods used (if any).[1]

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## 3.1 | Subsections and Subsubsections

This is a section.

### 3.1.1 | A Subsection

This is a subsection.

#### 3.1.1.1 | A Subsubsection

This is a subsubsection.

[1] For more information see: http://rc.rcjournal.com/content/49/10/1229.short. Last Accessed: 27 November 2024.

## 3.2 | Footnotes

Some text with a footnote, if an online link remember to add *Last Accessed*.[2]

## 3.3 | Equations

The following is the most beautiful equation in maths, Euler's Identity (Equation 3.1).

$$e^{i\pi} + 1 = 0 \tag{3.1}$$

where:

$e$ = the constant
$i$ = of complex fame
$\pi$ = not of the apple variety

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## 3.4 | Numbered Lists

This is an example of a numbered list:

1. This is my first point
2. My second
3. My third!
4. And my fourth?

[2] Some footnote text.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

## 3.5 | Bulleted Lists

This is an example of a bulleted list:

- This is my first point
- My second
- My third!
- And my fourth?

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## 3.6 | Figures

A test figure is shown in Figure 3.1.

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**Figure 3.1:** A test figure. This caption is huge, but in the list of figures only the smaller version in the square brackets will appear.

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## 3.7 | Two Side-by-Side Figures

Two figures side by side are shown in Figure 3.2.

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(a) Goku



(b) More Goku

**Figure 3.2:** The same super saiyan. Two times.

blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

## 3.8 | Acronyms

In the early nineties, GSM was deployed in many European countries. Global System for Mobile communication (GSM) offered for the first time international roaming for mobile subscribers. The GSM's use of Time Division Multiple Access (TDMA) as its communication standard was debated at length. And every now and then there are big discussions about whether Code Division Multiple Access (CDMA) should have been chosen over TDMA.

If you want to know more about Global System for Mobile communication (GSM), Time Division Multiple Access (TDMA), Code Division Multiple Access (CDMA) and other acronyms, just read a book about mobile communication. Just to mention it: There is another Used Acronym (UA), for testing.

## 3.9 | Tables

A beautiful table is shown in Table 3.1, data from Ebejer et al. (2012).

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**Table 3.1:** A Beautiful and Complex Table (table captions go above the table)

| | $w = 8$ | | | $w = 16$ | | |
|---|---|---|---|---|---|---|
| | $t = 0$ | $t = 1$ | $t = 2$ | $t = 0$ | $t = 1$ | $t = 2$ |
| $dir = 1$ | | | | | | |
| $c$ | 0.0790 | 0.1692 | 0.2945 | 0.3670 | 0.7187 | 3.1815 |
| $c$ | -0.8651 | 50.0476 | 5.9384 | -9.0714 | 297.0923 | 46.2143 |
| $c$ | 124.2756 | -50.9612 | -14.2721 | 128.2265 | -630.5455 | -381.0930 |
| $dir = 0$ | | | | | | |
| $c$ | 0.0357 | 1.2473 | 0.2119 | 0.3593 | -0.2755 | 2.1764 |
| $c$ | -17.9048 | -37.1111 | 8.8591 | -30.7381 | -9.5952 | -3.0000 |
| $c$ | 105.5518 | 232.1160 | -94.7351 | 100.2497 | 141.2778 | -259.7326 |

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## 3.10 | Long Tables

The following is an example of a table (Table 3.2) spanning multiple pages.

**Table 3.2:** Performance of Ligity in HTS mode against the Ligity-compatible DUD-E targets. The mean (and standard deviation in parentheses) values of ROC AUC using Tanimoto is 0.622 ($\pm$0.132), while for Tversky it is 0.671 ($\pm$0.142); the mean $EF_{1\%}$ using Tanimoto is 5.648 ($\pm$8.668), while for $EF_{1\%}$ using Tversky it is 9.047 ($\pm$12.713).

| Target | No. of Actives | No. of Decoys | ROC AUC Tanimoto | ROC AUC Tversky | BEDROC Tanimoto | BEDROC Tversky | $EF_{1\%}$ Tanimoto | $EF_{1\%}$ Tversky |
|---|---|---|---|---|---|---|---|---|
| ABL1 | 182 | 10,750 | 0.563 | 0.473 | 0.077 | 0.077 | 1.653 | 2.204 |
| ACE | 281 | 16,877 | 0.787 | 0.787 | 0.336 | 0.401 | 12.425 | 19.525 |
| ACES | 453 | 26,242 | 0.634 | 0.645 | 0.077 | 0.155 | 1.766 | 5.518 |
| ADA | 93 | 5,450 | 0.724 | 0.660 | 0.149 | 0.147 | 3.251 | 3.251 |
| ADA17 | 532 | 35,898 | 0.638 | 0.728 | 0.103 | 0.283 | 1.317 | 9.030 |
| ADRB1 | 247 | 15,850 | 0.523 | 0.647 | 0.065 | 0.129 | 1.619 | 5.262 |
| ADRB2 | 231 | 14,999 | 0.523 | 0.589 | 0.052 | 0.040 | 1.735 | 0.000 |
| AKT1 | 293 | 16,450 | 0.386 | 0.548 | 0.039 | 0.107 | 2.737 | 3.080 |
| AKT2 | 117 | 6,900 | 0.511 | 0.685 | 0.140 | 0.194 | 8.568 | 8.568 |

(continued...)

| Target | No. of Actives | No. of Decoys | ROC AUC Tanimoto | ROC AUC Tversky | BEDROC Tanimoto | BEDROC Tversky | $EF_{1\%}$ Tanimoto | $EF_{1\%}$ Tversky |
|---|---|---|---|---|---|---|---|---|
| ALDR | 159 | 8,988 | 0.574 | 0.610 | 0.202 | 0.172 | 10.747 | 6.322 |
| AMPC | 48 | 2,845 | 0.521 | 0.541 | 0.049 | 0.023 | 0.000 | 0.000 |
| ANDR | 269 | 14,349 | 0.722 | 0.742 | 0.194 | 0.354 | 4.839 | 24.938 |
| AOFB | 121 | 6,875 | 0.422 | 0.464 | 0.045 | 0.027 | 1.652 | 0.000 |
| BACE1 | 283 | 18,100 | 0.441 | 0.775 | 0.017 | 0.310 | 0.000 | 13.062 |
| BRAF | 152 | 9,950 | 0.612 | 0.639 | 0.208 | 0.165 | 12.502 | 5.264 |
| CASP3 | 199 | 10,694 | 0.600 | 0.734 | 0.068 | 0.258 | 0.502 | 7.031 |
| CDK2 | 474 | 27,838 | 0.467 | 0.507 | 0.021 | 0.048 | 0.000 | 1.055 |
| COMT | 41 | 3,846 | 0.789 | 0.889 | 0.338 | 0.665 | 19.447 | 58.341 |
| CP2C9 | 120 | 7,449 | 0.518 | 0.634 | 0.058 | 0.186 | 1.660 | 8.299 |
| CP3A4 | 170 | 11,787 | 0.450 | 0.493 | 0.022 | 0.057 | 0.000 | 2.345 |
| CSF1R | 166 | 12,149 | 0.526 | 0.542 | 0.136 | 0.152 | 6.031 | 7.238 |
| CXCR4 | 40 | 3,405 | 0.575 | 0.722 | 0.217 | 0.134 | 12.665 | 0.000 |
| DEF | 102 | 5,699 | 0.732 | 0.833 | 0.212 | 0.379 | 10.786 | 15.689 |
| DHI1 | 330 | 19,348 | 0.481 | 0.595 | 0.089 | 0.062 | 2.422 | 1.211 |
| DPP4 | 533 | 40,941 | 0.586 | 0.591 | 0.154 | 0.157 | 4.312 | 3.937 |
| DRD3 | 480 | 34,048 | 0.484 | 0.441 | 0.043 | 0.046 | 1.251 | 0.626 |
| DYR | 231 | 17,196 | 0.694 | 0.758 | 0.210 | 0.230 | 6.504 | 7.371 |
| EGFR | 542 | 35,047 | 0.593 | 0.491 | 0.054 | 0.037 | 0.922 | 0.000 |
| ESR1 | 383 | 20,683 | 0.838 | 0.861 | 0.527 | 0.594 | 31.281 | 39.101 |
| ESR2 | 367 | 20,199 | 0.844 | 0.870 | 0.563 | 0.644 | 20.130 | 32.644 |
| FA10 | 537 | 28,324 | 0.564 | 0.674 | 0.058 | 0.118 | 0.930 | 2.232 |
| FA7 | 114 | 6,249 | 0.762 | 0.859 | 0.210 | 0.332 | 6.105 | 8.721 |
| FABP4 | 47 | 2,749 | 0.786 | 0.744 | 0.191 | 0.276 | 0.000 | 10.623 |
| FAK1 | 100 | 5,350 | 0.642 | 0.531 | 0.111 | 0.065 | 2.019 | 0.000 |
| FGFR1 | 139 | 8,698 | 0.511 | 0.522 | 0.036 | 0.088 | 0.722 | 1.445 |
| FKB1A | 111 | 5,799 | 0.605 | 0.751 | 0.162 | 0.164 | 8.122 | 3.610 |
| FNTA | 592 | 51,493 | 0.411 | 0.625 | 0.012 | 0.132 | 0.000 | 4.053 |
| FPPS | 85 | 8,842 | 0.917 | 0.985 | 0.323 | 0.776 | 2.360 | 36.581 |
| GCR | 258 | 14,998 | 0.805 | 0.834 | 0.244 | 0.324 | 3.092 | 8.116 |
| GLCM | 54 | 3,790 | 0.667 | 0.685 | 0.182 | 0.279 | 1.873 | 11.240 |
| GRIA2 | 158 | 11,842 | 0.662 | 0.684 | 0.248 | 0.154 | 11.392 | 5.696 |
| GRIK1 | 101 | 6,547 | 0.656 | 0.668 | 0.203 | 0.102 | 7.978 | 1.995 |
| HDAC2 | 185 | 10,300 | 0.676 | 0.734 | 0.187 | 0.201 | 4.318 | 4.318 |
| HDAC8 | 170 | 10,449 | 0.640 | 0.819 | 0.120 | 0.377 | 2.946 | 8.250 |
| HIVINT | 100 | 6,640 | 0.390 | 0.554 | 0.030 | 0.116 | 0.000 | 3.018 |
| HIVPR | 535 | 35,724 | 0.663 | 0.872 | 0.072 | 0.490 | 0.187 | 23.898 |
| HIVRT | 338 | 18,884 | 0.495 | 0.475 | 0.124 | 0.085 | 4.443 | 1.777 |
| HMDH | 170 | 8,750 | 0.480 | 0.906 | 0.068 | 0.652 | 2.358 | 35.963 |
| HS90A | 88 | 4,850 | 0.635 | 0.506 | 0.096 | 0.083 | 0.000 | 3.436 |
| HXK4 | 92 | 4,700 | 0.662 | 0.803 | 0.206 | 0.307 | 15.192 | 9.766 |
| IGF1R | 148 | 9,300 | 0.502 | 0.575 | 0.057 | 0.189 | 2.037 | 14.941 |
| INHA | 43 | 2,300 | 0.493 | 0.575 | 0.031 | 0.045 | 0.000 | 0.000 |
| ITAL | 138 | 8,500 | 0.619 | 0.465 | 0.037 | 0.065 | 0.000 | 0.728 |
| JAK2 | 107 | 6,500 | 0.472 | 0.475 | 0.073 | 0.118 | 2.807 | 6.549 |
| KIF11 | 116 | 6,850 | 0.755 | 0.781 | 0.149 | 0.219 | 4.289 | 2.574 |
| KIT | 166 | 10,449 | 0.463 | 0.437 | 0.045 | 0.030 | 0.000 | 0.000 |
| KITH | 57 | 2,850 | 0.649 | 0.838 | 0.228 | 0.709 | 14.069 | 47.483 |
| KPCB | 135 | 8,699 | 0.753 | 0.813 | 0.220 | 0.338 | 8.923 | 12.641 |
| LCK | 419 | 27,391 | 0.471 | 0.437 | 0.031 | 0.043 | 0.000 | 1.910 |
| LKHA4 | 171 | 9,448 | 0.718 | 0.694 | 0.238 | 0.150 | 8.203 | 1.758 |
| MAPK2 | 101 | 6,148 | 0.660 | 0.670 | 0.174 | 0.199 | 5.988 | 3.992 |
| MCR | 94 | 5,149 | 0.816 | 0.888 | 0.215 | 0.454 | 6.436 | 19.307 |
| MET | 166 | 11,249 | 0.566 | 0.531 | 0.130 | 0.065 | 6.032 | 0.603 |

(continued...)

| Target | No. of Actives | No. of Decoys | ROC AUC Tanimoto | ROC AUC Tversky | BEDROC Tanimoto | BEDROC Tversky | $EF_{1\%}$ Tanimoto | $EF_{1\%}$ Tversky |
|---|---|---|---|---|---|---|---|---|
| MK01 | 79 | 4,550 | 0.518 | 0.602 | 0.121 | 0.206 | 5.095 | 3.821 |
| MK10 | 104 | 6,600 | 0.488 | 0.489 | 0.020 | 0.031 | 0.962 | 0.962 |
| MK14 | 578 | 35,847 | 0.511 | 0.589 | 0.040 | 0.064 | 0.173 | 0.519 |
| MMP13 | 572 | 37,199 | 0.648 | 0.753 | 0.134 | 0.268 | 2.446 | 9.957 |
| MP2K1 | 121 | 8,146 | 0.669 | 0.569 | 0.187 | 0.058 | 3.293 | 0.823 |
| NOS1 | 98 | 8,028 | 0.483 | 0.451 | 0.109 | 0.041 | 3.071 | 0.000 |
| NRAM | 98 | 6,200 | 0.853 | 0.859 | 0.342 | 0.290 | 11.221 | 3.060 |
| PA2GA | 99 | 5,150 | 0.793 | 0.756 | 0.225 | 0.153 | 1.020 | 3.059 |
| PARP1 | 508 | 30,029 | 0.635 | 0.692 | 0.215 | 0.231 | 11.234 | 7.884 |
| PGH1 | 195 | 10,798 | 0.645 | 0.637 | 0.077 | 0.100 | 0.000 | 2.050 |
| PGH2 | 435 | 23,139 | 0.716 | 0.780 | 0.166 | 0.291 | 3.444 | 9.874 |
| PLK1 | 107 | 6,800 | 0.658 | 0.531 | 0.123 | 0.048 | 1.871 | 0.000 |
| PNPH | 103 | 6,946 | 0.575 | 0.578 | 0.161 | 0.181 | 4.888 | 8.799 |
| PPARA | 373 | 19,399 | 0.783 | 0.778 | 0.262 | 0.280 | 6.693 | 7.764 |
| PPARD | 240 | 12,250 | 0.547 | 0.544 | 0.078 | 0.098 | 1.665 | 2.498 |
| PPARG | 484 | 25,299 | 0.515 | 0.605 | 0.055 | 0.118 | 0.619 | 4.955 |
| PRGR | 293 | 15,648 | 0.740 | 0.793 | 0.142 | 0.318 | 2.053 | 14.714 |
| PTN1 | 130 | 7,249 | 0.398 | 0.538 | 0.055 | 0.090 | 0.000 | 3.068 |
| PUR2 | 50 | 2,700 | 0.851 | 0.837 | 0.281 | 0.255 | 7.857 | 1.964 |
| PYGM | 77 | 3,944 | 0.403 | 0.492 | 0.016 | 0.137 | 0.000 | 3.917 |
| PYRD | 111 | 6,449 | 0.682 | 0.710 | 0.462 | 0.413 | 34.027 | 16.118 |
| RENI | 104 | 6,956 | 0.720 | 0.789 | 0.043 | 0.138 | 0.000 | 0.000 |
| ROCK1 | 100 | 6,300 | 0.347 | 0.449 | 0.020 | 0.084 | 1.000 | 4.000 |
| RXRA | 131 | 6,950 | 0.788 | 0.900 | 0.219 | 0.596 | 6.091 | 27.407 |
| SAHH | 63 | 3,450 | 0.874 | 0.852 | 0.598 | 0.542 | 35.050 | 27.084 |
| SRC | 524 | 34,500 | 0.565 | 0.477 | 0.065 | 0.050 | 0.382 | 0.573 |
| TGFR1 | 133 | 8,499 | 0.609 | 0.639 | 0.147 | 0.154 | 10.565 | 4.528 |
| THB | 103 | 7,450 | 0.794 | 0.762 | 0.238 | 0.150 | 10.614 | 0.965 |
| THRB | 461 | 27,000 | 0.605 | 0.706 | 0.063 | 0.166 | 2.166 | 5.632 |
| TRY1 | 449 | 25,975 | 0.711 | 0.815 | 0.147 | 0.280 | 2.898 | 6.688 |
| TRYB1 | 148 | 7,650 | 0.670 | 0.670 | 0.153 | 0.132 | 3.378 | 3.378 |
| TYSY | 109 | 6,745 | 0.594 | 0.725 | 0.071 | 0.226 | 0.911 | 5.468 |
| UROK | 162 | 9,850 | 0.525 | 0.650 | 0.036 | 0.120 | 0.000 | 1.854 |
| VGFR2 | 409 | 24,948 | 0.632 | 0.578 | 0.083 | 0.093 | 1.465 | 1.465 |
| WEE1 | 102 | 6,150 | 0.934 | 0.929 | 0.789 | 0.797 | 59.348 | 61.294 |
| XIAP | 100 | 5,150 | 0.752 | 0.974 | 0.190 | 0.897 | 8.077 | 51.490 |

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## 3.11 | Landscape Tables

Next is an example of a wide table on a landscape oriented paper (Table 3.3).

**Table 3.3:** A table in landscape orientation.

| $m$ | $x$ | $y$ | $z$ | $a$ | $A_m$ | $B$ | $C$ | $x$ | $y$ | $z$ | $a$ | $A_m$ | $B$ | $C$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 16.128 | +8.872 | 16.128 | 1.402 | 1.373 | -146.6 | -137.6 | 16.128 | +8.872 | 16.128 | 1.402 | 1.373 | -146.6 | -137.6 |
| 2 | 3.442 | -2.509 | 3.442 | 0.299 | 0.343 | 133.2 | 152.4 | 3.442 | -2.509 | 3.442 | 0.299 | 0.343 | 133.2 | 152.4 |
| 3 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 4 | 0.993 | -0.429 | 0.993 | 0.086 | 0.08 | 25.6 | 90 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 5 | 1.29 | +0.099 | 1.29 | 0.112 | 0.097 | -175.6 | -114.7 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 6 | 0.483 | -0.183 | 0.483 | 0.042 | 0.063 | 22.3 | 122.5 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 7 | 0.766 | -0.475 | 0.766 | 0.067 | 0.039 | 141.6 | -122 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 8 | 0.624 | +0.365 | 0.624 | 0.054 | 0.04 | -35.7 | 90 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 9 | 0.641 | -0.466 | 0.641 | 0.056 | 0.045 | 133.3 | -106.3 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 10 | 0.45 | +0.421 | 0.45 | 0.039 | 0.034 | -69.4 | 110.9 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |
| 11 | 0.598 | -0.597 | 0.598 | 0.052 | 0.025 | 92.3 | -109.3 | 1.826 | -0.363 | 1.826 | 0.159 | 0.119 | 168.5 | -161.1 |

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## 3.12 | Theorems

**Theorem 3.12.1.** *Let $f$ be a function whose derivative exists in every point, then $f$ is a continuous function.*

**Theorem 3.12.2** (Pythagorean theorem)**.** *This is a theorem about right triangles and can be summarised in the next equation*

$$x^2 + y^2 = z^2$$

And a consequence of Theorem 3.12.2 is the statement in the next corollary.

**Corollary 3.12.2.1.** *There's no right triangle whose sides measure 3 cm, 4 cm, and 6 cm.*

You can reference theorems, such as Theorem 3.12.2, when a label is assigned.

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## 3.13 | Lemmas

**Lemma 3.13.1.** *Given two line segments whose lengths are $a$ and $b$ respectively there is a real number $r$ such that $b = ra$.*

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## 3.14 | Proofs

**Lemma 3.14.1.** *Given two line segments whose lengths are $a$ and $b$ respectively there is a real number $r$ such that $b = ra$.*

*Proof.* To prove it by contradiction try and assume that the statement is false, proceed from there and at some point you will arrive at a contradiction. □

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## 3.15 | Code Listings

Here you go.

```
import numpy as np

def incmatrix(genl1, genl2):
    m = len(genl1)
    n = len(genl2)
    M = None  # to become the incidence matrix
    VT = np.zeros((n * m, 1), int)  # dummy variable
```

```
8
9     # compute the bitwise xor matrix
10    M1 = bitxormatrix(genl1)
11    M2 = np.triu(bitxormatrix(genl2), 1)
12
13    for i in range(m - 1):
14        for j in range(i + 1, m):
15            [r, c] = np.where(M2 == M1[i, j])
16            for k in range(len(r)):
17                VT[i * n + r[k]] = 1
18                VT[i * n + c[k]] = 1
19                VT[j * n + r[k]] = 1
20                VT[j * n + c[k]] = 1
21
22            if M is None:
23                M = np.copy(VT)
24            else:
25                M = np.concatenate((M, VT), 1)
26
27            VT = np.zeros((n * m, 1), int)
28
29    return M
```

**Listing 3.1:** My Listing Caption

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## 3.16 | Algorithms

An algorithm example is shown in Algorithm 1. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum

**Algorithm 1:** An algorithm with caption

```
Data: n ≥ 0
Result: y = x^n
y ← 1;
X ← x;
N ← n;
while N ≠ 0 do
    if N is even then
        X ← X × X;
        N ← N/2 ;                          /* This is a comment */
    else
        if N is odd then
            y ← y × X;
            N ← N − 1;
        end
    end
end
```

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## 3.17 | Suppressing Page Numbers on a Float Page

Kindly refer to Figure 3.3.

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**Figure 3.3:** Page numbers are suppressed on this page.

## 3.18 | Referencing

Use `\textcite` for in-text citations, e.g. Einstein (1905), and `\parencite` for citations in parentheses. And this is what an online reference looks like (LLC, 1999).

In their study, Einstein (1905) show the world is round. Others have shown this to be the case (Arrighi, 2003; Ebejer et al., 2016).

## 3.19 | Using a Different Language

Local name = Malti. Jiena ngħix Ħaż-Żebbuġ.

## 3.20 | Summary

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# 4

# Results & Discussion

Should include a reiteration of the experiments, and their outcome. Together with a description (discussion). Preamble should include a reminder of the aims and objectives together with a list of experiments to achieve these. Should include many charts and other visualisations with appropriate descriptions.[1]

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## 4.1 | Summary

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[1] Another footnote example.

# 5

# Evaluation

In an ideal world, you should have two kinds of evaluation. The first is against some ground truth (perhaps a random model?). The second kind of evaluation is against other people's work (accuracy, speed, etc.). Any dimension of interest should be evaluated. Evaluation should be statistically sound.

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## 5.1 | Summary

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# 6

# Conclusion

This section should have a summary of the whole project. The original aims and objectives and whether these have been met should be discussed. It should include a section with a critique and a list of limitations of your proposed solutions. Future work should be described, and this should not be marginal or silly (e.g. add machine learning models). It is always good to end on a positive note (i.e. 'Final Remarks').

## 6.1 | Revisiting the Aims and Objectives

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## 6.2 | Main Contributions

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## 6.3 | Critique and Limitations

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## 6.4 | Future Work

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## 6.5 | Final Remarks

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# A

# Media Content

If the dissertation has a DVD or pendrive attached to it, you will need a section which explains what is on the media (structure, files, data, etc.). This could be a table with filename and description.

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# B

# Installation Instructions

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# C

# User Manual

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